InfoGAN: Interpretable Representation Learning by Information Maximizing Generative Adversarial Nets

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Introduction & Motivation

Problem

- GANs generate realistic samples
- Latent space is entangled and non-interpretable

Goal

- Learn disentangled, interpretable representations
- Fully unsupervised

Solution

- InfoGAN: GAN + information maximization

GAN Background

GAN framework

Generator G(z)G(z)G(z)

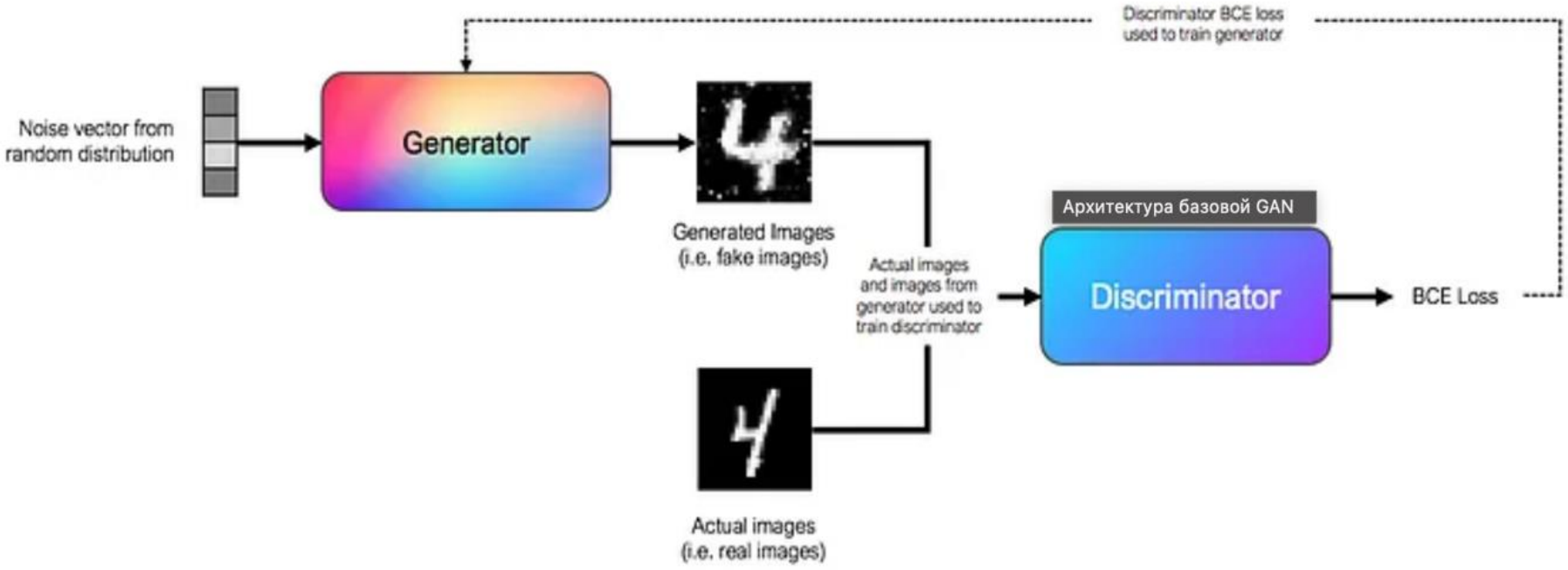
Discriminator D(x)D(x)D(x)

Training

Minimax adversarial game

Limitation

No control over semantics of latent variables



Main Goal

- Develop a generative model that performs unsupervised disentanglement.
- Automatically discover interpretable factors of variation in latent space.
- Enable precise, independent control over specific attributes in generated images.
- Achieve this without using labels during training.

Core Idea of InfoGAN

Split latent input:

z: noise

c: latent code with semantic meaning

- Categorical (c_cat) → discrete factors (digit class).
- Continuous (c_cont) → smooth variations (stroke thickness, rotation).

Introduces auxiliary **Q-network** on top of the discriminator to approximates p(c|x)

and makes mutual information tractable.

Mutual Information (MI) maximization: encourages generator to align images with latent codes. I(c;G(z,c))

Result

Latent code controls interpretable variations

InfoGAN Architecture

Generator: Fully connected + deconvolutional layers, input is [z, c_cat, c_cont].

Discriminator: Convolutional network producing real/fake score.

Q-network: Shares features with discriminator, predicts latent codes for MI maximization.

Hyperparameters:

$z \in R^{62}$, c_cat = 10, c_cont = 2

λ_MI = 1.0, Adam optimizer $\beta 1$=0.5, lr=2e-4

Experimental Setup

Model

GAN + auxiliary **Q-network**

Q shares convolutional layers with discriminator

Latent codes

Noise: $z \sim N(0,I)$

Structured code ccc:

- Categorical (e.g. 10 classes)
- Continuous (uniform in [−1,1][-1,1][−1,1])

Training objective

Standard GAN loss

- mutual information lower bound

$E[\log Q(c|G(z,c))]$

Results on MNIST

Setup

- 1 categorical code (10 values)
- 2 continuous codes

Observed disentanglement

- Categorical code → digit identity
- Continuous codes →
  rotation
  stroke thickness

Key result

No labels used during training

Realization: Dataset

MNIST (handwritten digits)

- Training set: 60,000 images
- Normalized to [-1, 1]
- Grayscale 28×28 pixels
- Fully unsupervised training: **no labels used during GAN training**

Realization: Original InfoGAN (Chen et al., 2016)

```
z (62)   c_cat (10)   c_cont (2)
   \        |          /
    \_______|__________/
            |
        Generator
    (FC + Deconv layers)
            |
      Generated Image (28×28)
            |
      Discriminator
   (Conv + FC → Real/Fake)
            |
      Auxiliary Q-network
   (Predict c_cat + c_cont)
```

- Generator: fully connected + deconv
- Discriminator: conv + fc
- Q-network: predicts categorical and continuous codes
- $z \sim N(0,1)$, c_cat = 10 categories, c_cont = 2
- Mutual information maximization via Q-network
- Optimizer: Adam, $\beta 1 = 0.5$, lr = 2e−4

Realization: InfoGAN in my code

```
z (62)   c_cat (10)   c_cont (2)
   \        |          /
    \_______|__________/
            |
     Generator (PyTorch)
FC: 74 → 1024 → 128*7*7 → reshape → (128,7,7)
Deconv: 128→64→1 + BN + ReLU/Tanh
            |
      Generated Image (28×28)
            |
     Discriminator (PyTorch)
Conv + FC → Real/Fake
            |
      Q-network (on top of D)
Predict c_cat + c_cont
```

Генератор: fully connected + deconv (PyTorch)
Дискриминатор: conv + fc, LeakyReLU + BatchNorm
Q-сеть: использует shared features дискриминатора
$z \sim N(0,1)$, c_cat = 10, c_cont = 2, full vector = 74
Loss: BCE для D/G, MI loss для Q (λ_MI=1.0)
Optimizer: Adam, $\beta 1=0.5$, lr=2e−4

Realization: My Implementation Details

- **Shared features**: Q-network reuses D's 1024-dim features → efficient, as in paper
- **Continuous code variance**: exp(fc_var) to ensure positivity
- **Stability**: LeakyReLU in D, BatchNorm in G/D → standard DCGAN trick
- **Mutual Information weighting**: λ_MI=1.0, as in the appendix of the paper

Realization: Vanilla GAN for Comparison

```
z (62)
  |
Generator (PyTorch)
FC: 62 → 1024 → 128*7*7 → reshape → (128,7,7)
Deconv: 128→64→1 + BN + ReLU/Tanh
  |
Generated Image (28×28)
  |
Discriminator (PyTorch)
Conv + FC → Real/Fake
```

- Generator: same architecture as InfoGAN, but **no structured latent codes**
- Discriminator: same as InfoGAN
- No Q-network, no mutual information loss
- Input: only $z \sim N(0,1)$
- Loss: BCE for D/G
- Optimizer: Adam, $\beta 1 = 0.5$, lr = 2e−4

Evaluation & Metrics: Quantitative Evaluation

Metric	How it is computed	Purpose
Category Consistency Accuracy	Use pretrained MNIST classifier. Generate 500 images per categorical code ccat=kc_{cat}=kccat=k, predict digit labels, compute fraction matching kkk	Measures how well the categorical latent code controls digit identity
Macro F1-score	Same as above, but computes **macro-averaged F1** over all 10 digit classes	Balanced metric capturing both precision and recall; proxy for disentanglement quality
FID (optional)	Generate images, extract Inception-V3 features, compute Fréchet distance with real MNIST features	Standard generative model metric; captures realism and diversity

Evaluation & Metrics: Qualitative Evaluation

Latent Traversal

- Vary one latent variable (categorical or continuous) while keeping others fixed
- Observe generated images to see interpretable changes:
  - ccatc_{cat}ccat → digit identity
  - ccontc_{cont}ccont → stroke thickness, slant, style

Provides visual confirmation of disentanglement

Quantitative Results

Model	Category Accuracy ↑	Macro F1 ↑
Vanilla GAN	0.1	0.1
InfoGAN	0.78	0.73

InfoGAN clearly outperforms Vanilla GAN in aligning latent codes with digit identity.

- Accuracy and F1-score confirm disentanglement and interpretability.
- Original paper (Chen et al., 2016) shows only qualitative results; our metrics provide numeric validation.
- Mutual information maximization effectively enforces meaningful latent structure.

Conclusion

We implemented the InfoGAN framework on MNIST using PyTorch, faithfully reproducing the original TensorFlow architecture. We also trained a vanilla GAN baseline for comparison. Our experiments show that Vanilla GAN performs near-randomly and cannot map latent codes to digits, while InfoGAN achieves high category accuracy and macro F1-score. These results confirm that mutual information maximization effectively enforces disentangled and interpretable latent factors.